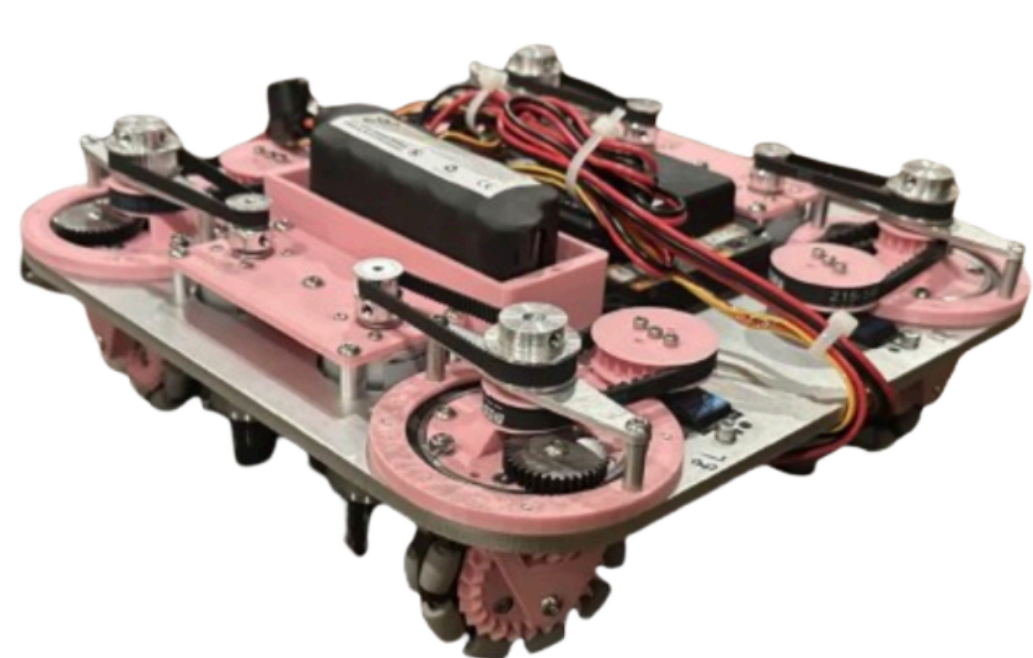
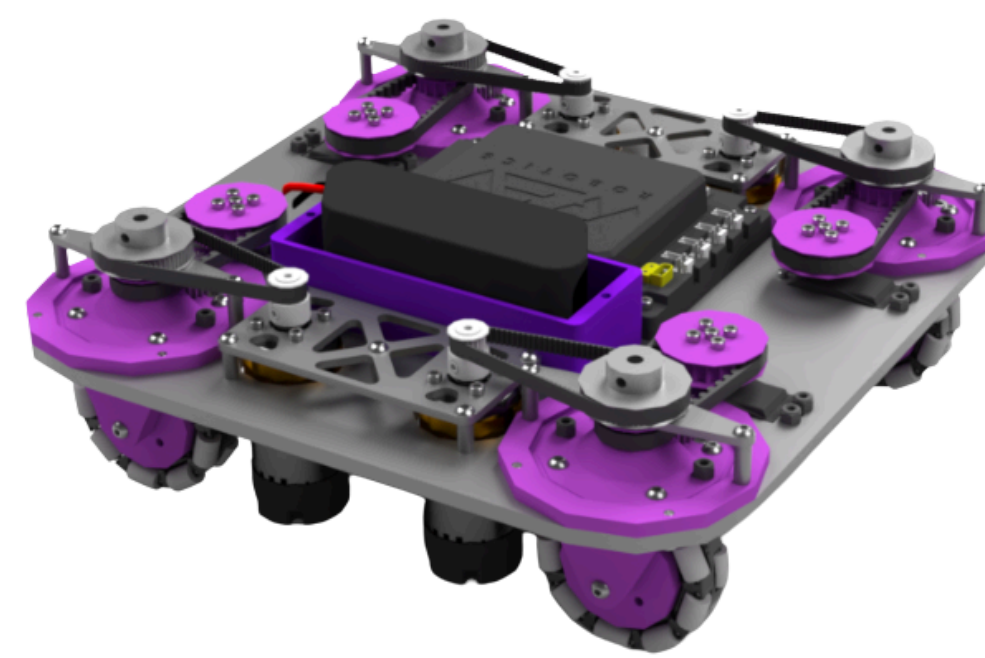




Optimization of Kinematics and Control of a Coaxial Swerve Omni Based Drive

Engineering: Electrical, Mechanical, and Robotics

Rahul Verma, Grade 11

BACKGROUND

"Swerve" drives utilize wheels that rotate on yaw and pitch axes for holonomic motion. These drivetrains can be beneficial in robotic applications requiring full power four directional movement. However, the effects of implementing an *omni-based* Swerve drive has not been researched.

In a standard, non-swerve, 4-wheel drive, a drivetrain max speed is limited by a preset gear ratio. Traditional cars have separate mechanisms such as continuously variable transmissions (CVT) to allow for variable gear ratios. However, an omni based swerve drive can achieve a CVT based on the rotation of the wheels, while also achieving holonomic motion. This is because the omni wheels have an unconstrained degree of freedom perpendicular to the wheel rotation, allowing for the wheel to have an artificially higher wheel diameter until, theoretically, infinity-- or, practically, when frictional losses take over.

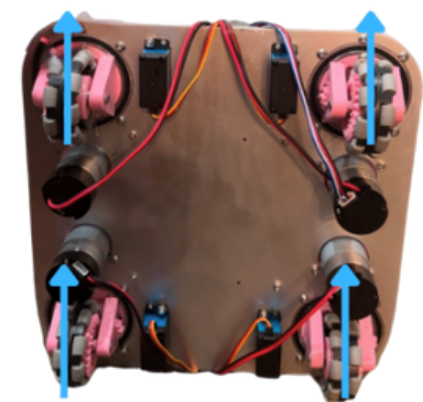


Figure 1: Modules Straight Forward

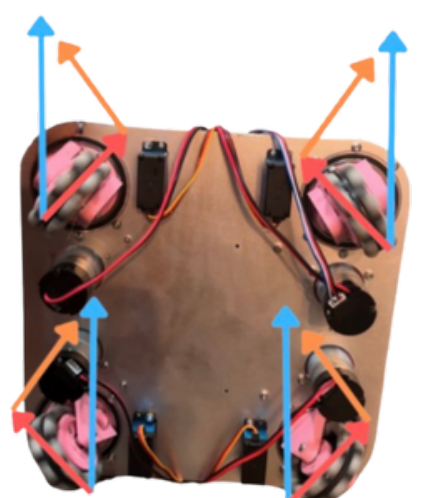


Figure 2: Modules Angled Inwards (CVT)

PROBLEM AND HYPOTHESIS

How does the continuously variable transmission (CVT) effect in an omni-based swerve drive impact travel time, and what is the optimal speed percentage for dynamic gear ratio adjustments to maximize efficiency?

The hypothesis is that a velocity-based control strategy, where the gear ratio decreases at 50% of top speed, will result in the lowest travel times.

CRITERIA/CONSTRAINTS

The prototype will optimize the control of a coaxial swerve omni-based drive to improve acceleration and minimize travel times. The primary criteria for success will be achieving a faster travel time while maintaining stability. This will be tested by measuring time traveled and acceleration profiles under varying percentages of top speed required before increasing the gear ratio. Constraints include the physical limitations of available motors, sensors and servos. Additionally, the design must account for mechanical constraints such as friction and gear backlash.

MATERIALS

Structural Materials:

- ¼" 6061 Aluminum Center Plate, laser cut
- ½" 6061 Aluminum Bearing Plate, laser cut
- 3D Printed parts in standard PLA

Mechanical Components:

- 4x Omni Wheels
- Assorted Bearings
- Aluminum Shafting
- Mod 0.8 Pinion Gears
- GT2 & HTD5 Belts

Hardware:

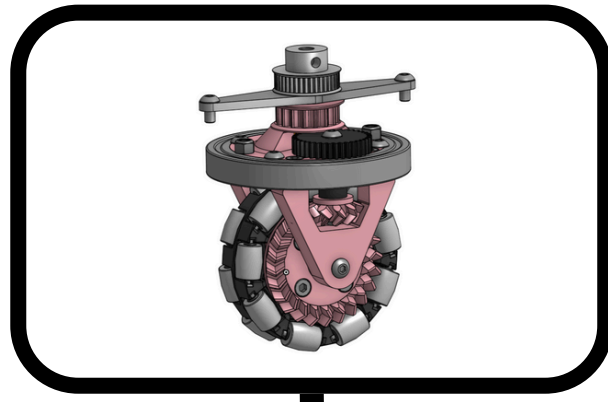
- M4 Socket screws
- M4 Button screws
- 6mm drill bit
- M4 Nylock nuts
- M3 Socket screws
- M3 Nylock nuts
- M4 Set Screws

Electronics:

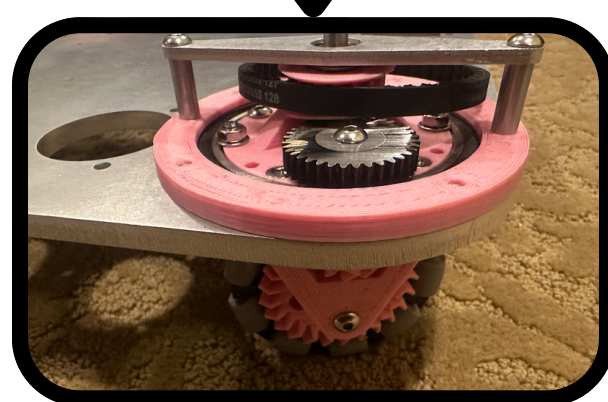
- Android Phone (for control)
- REV Robotics Expansion Hub
- Motor Encoder & Accelerometer
- 12V DC Brushed Motor
- 35kg DS3225 servos

DESIGN EXECUTION

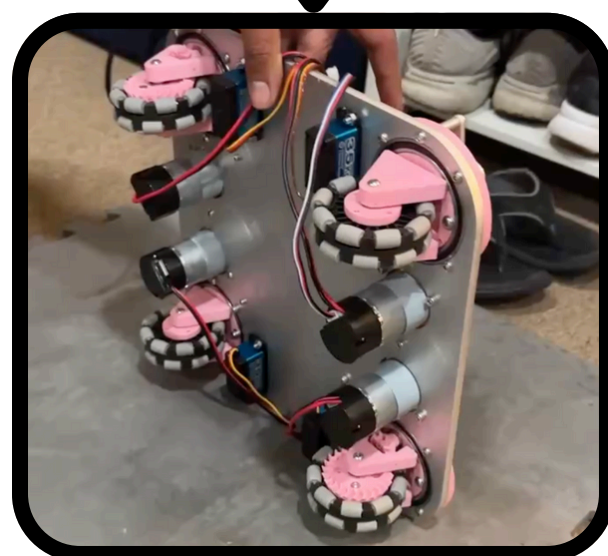
1. CAD & Prototype Design
 - a. Used CAD to design four wheel omni swerve
 - b. Iterated through 3D Printing & testing of pods individually
2. Fabrication & Assembly
 - a. Printed using FDM printer
 - b. Wired and integrated electronics
3. Software Implementation
 - a. Implemented kinematics to facilitate dynamic control of the desired gear ratio
 - b. Wrote testing programs to verify mechanical behavior
4. Testing & Data Collection
 - a. In each trial, the robot reached a different percent of top speed before the gear ratio was decreased in accordance with the kinematics to increase speed and decrease torque)
 - b. Time taken to travel 12 & 18ft was measured, with 5 trials for each percent of robot velocity



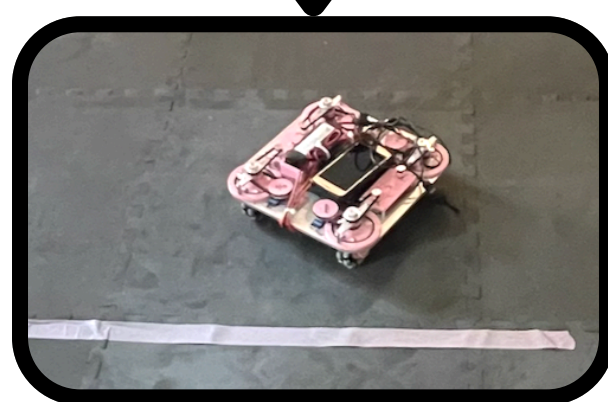
Design in CAD



Assemble Modules

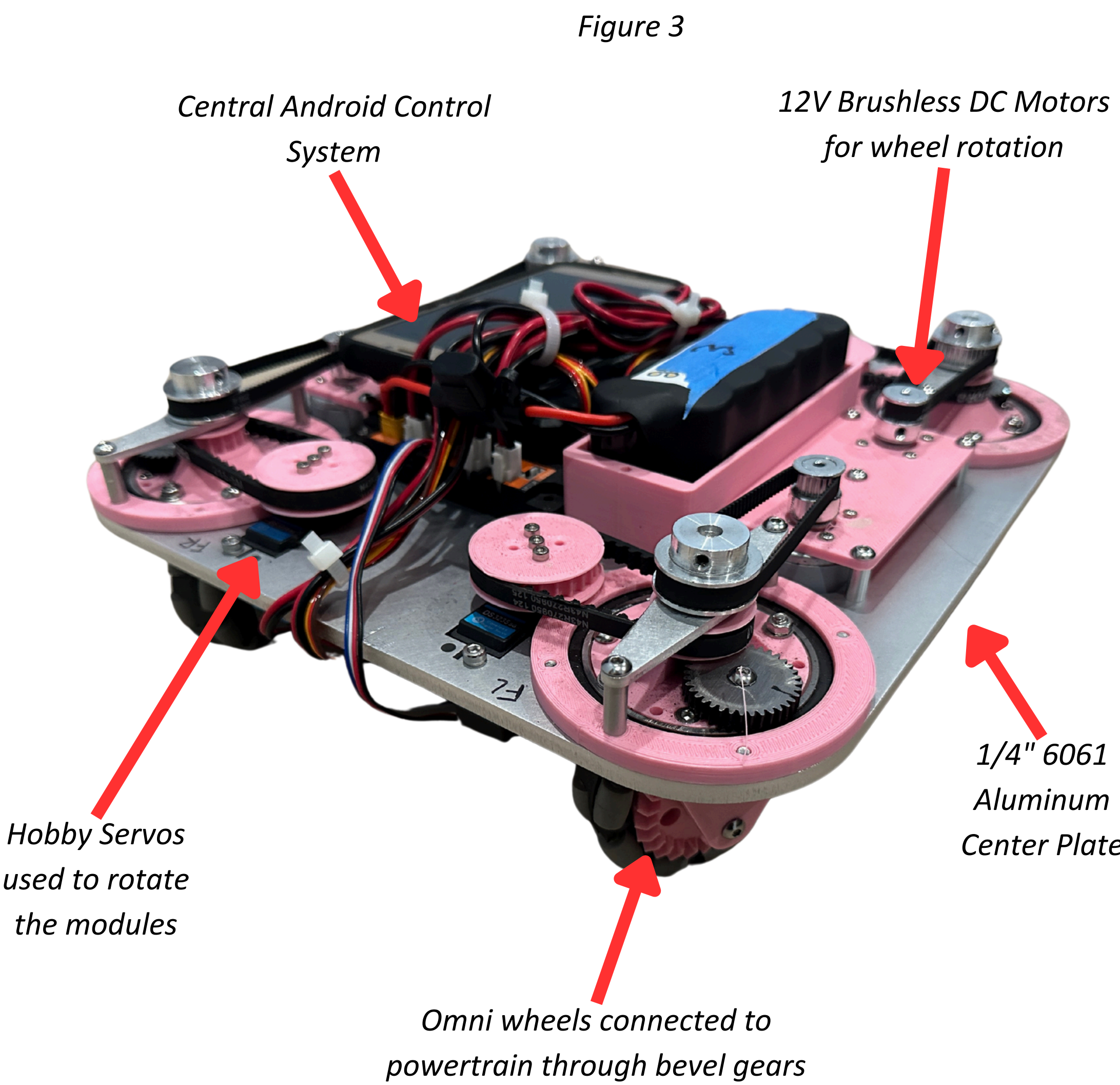


Fine Tune Software



Perform Experiments

FINAL DESIGN



RESULTS

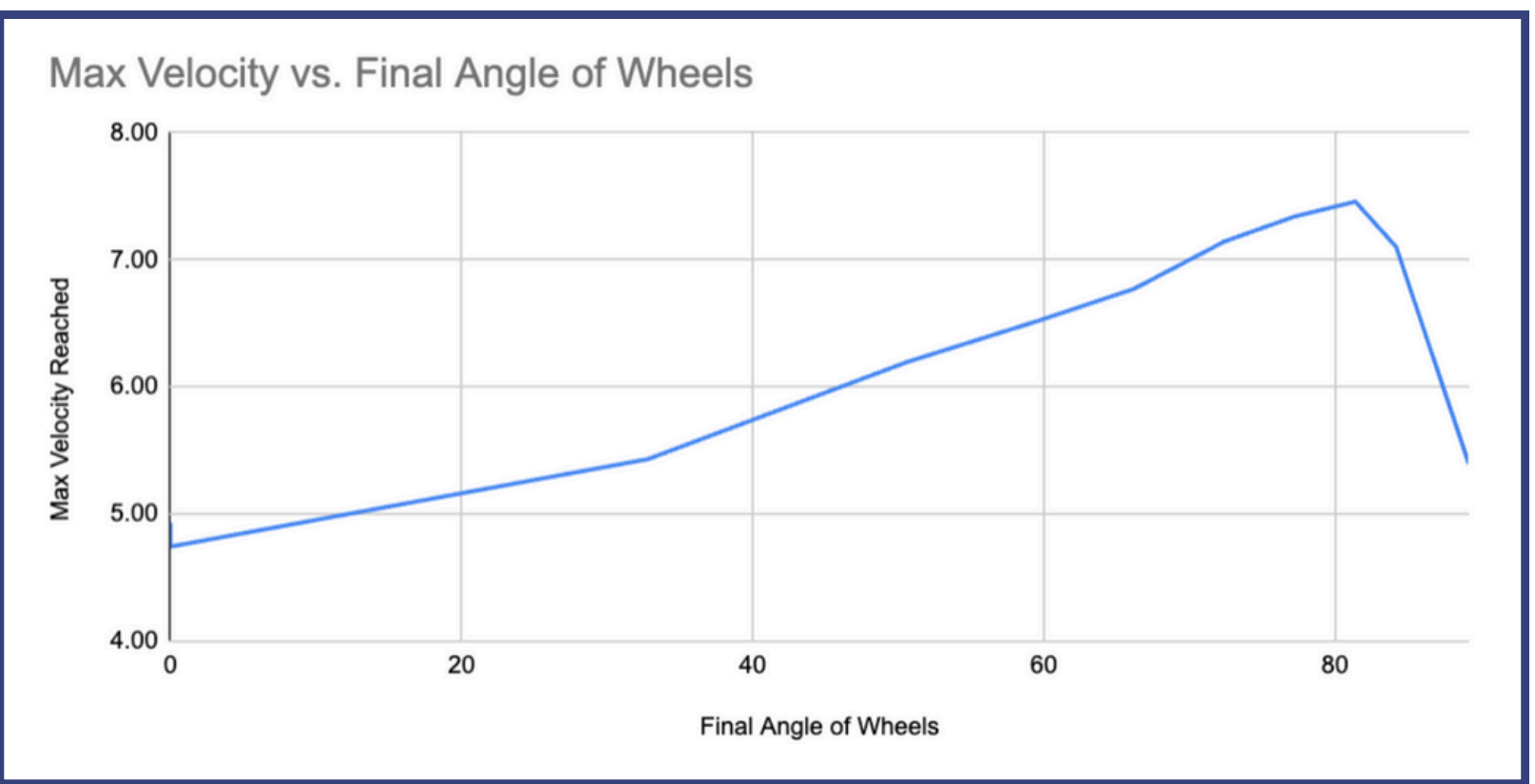


Figure 4: Velocity as a Function of Wheel Angle

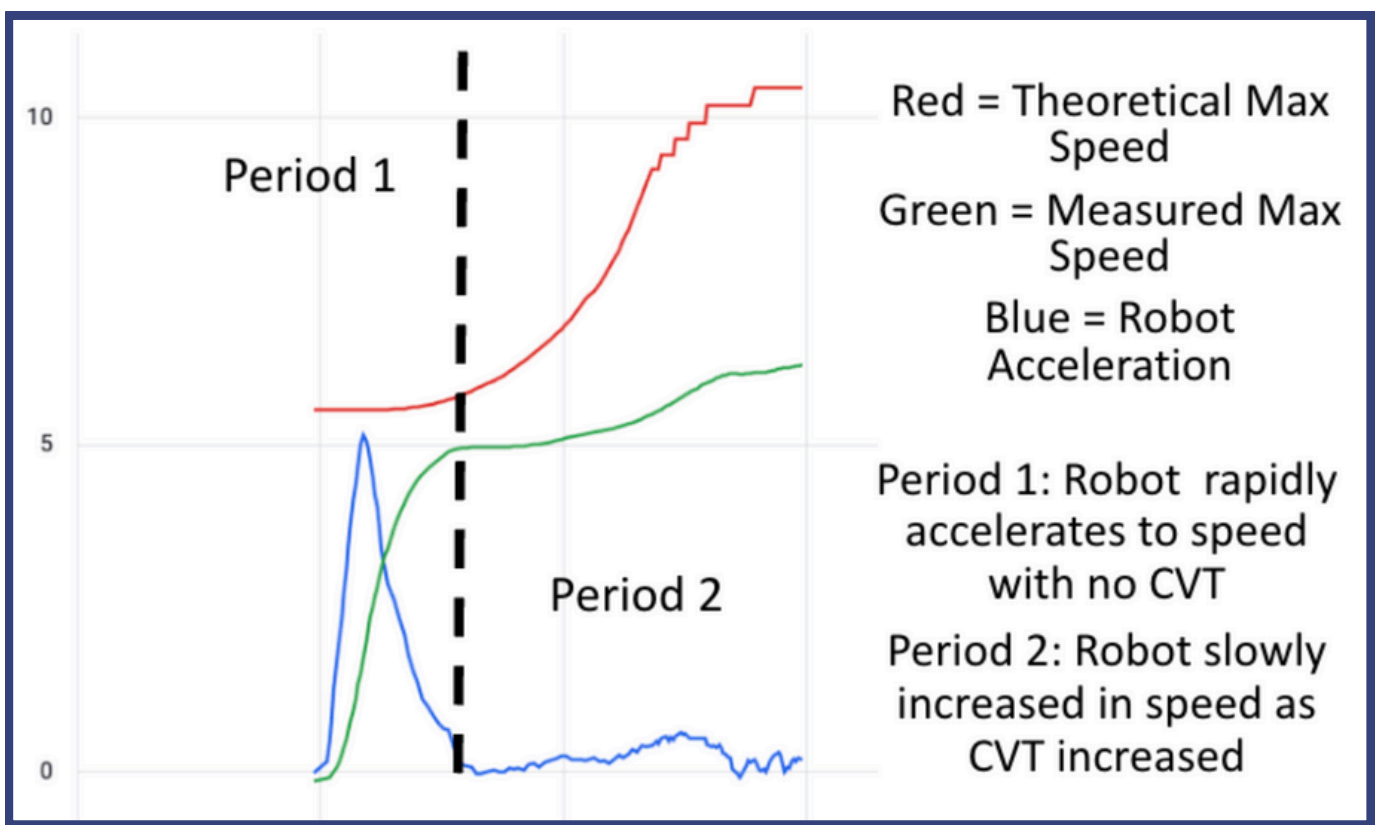


Figure 5: Acceleration and Velocity Profile

12 Feet			
Max Speed %	Avg Time Taken	Avg Max Speed	
0	2.64	5.39	
0.1	2.58	7.10	
0.2	2.52	7.46	
0.3	2.48	7.34	
0.4	2.40	7.14	
0.5	2.45	6.77	
0.6	2.47	6.52	
0.7	2.50	6.20	
0.8	2.60	5.43	
0.9	2.70	4.74	
1	2.654	4.932	

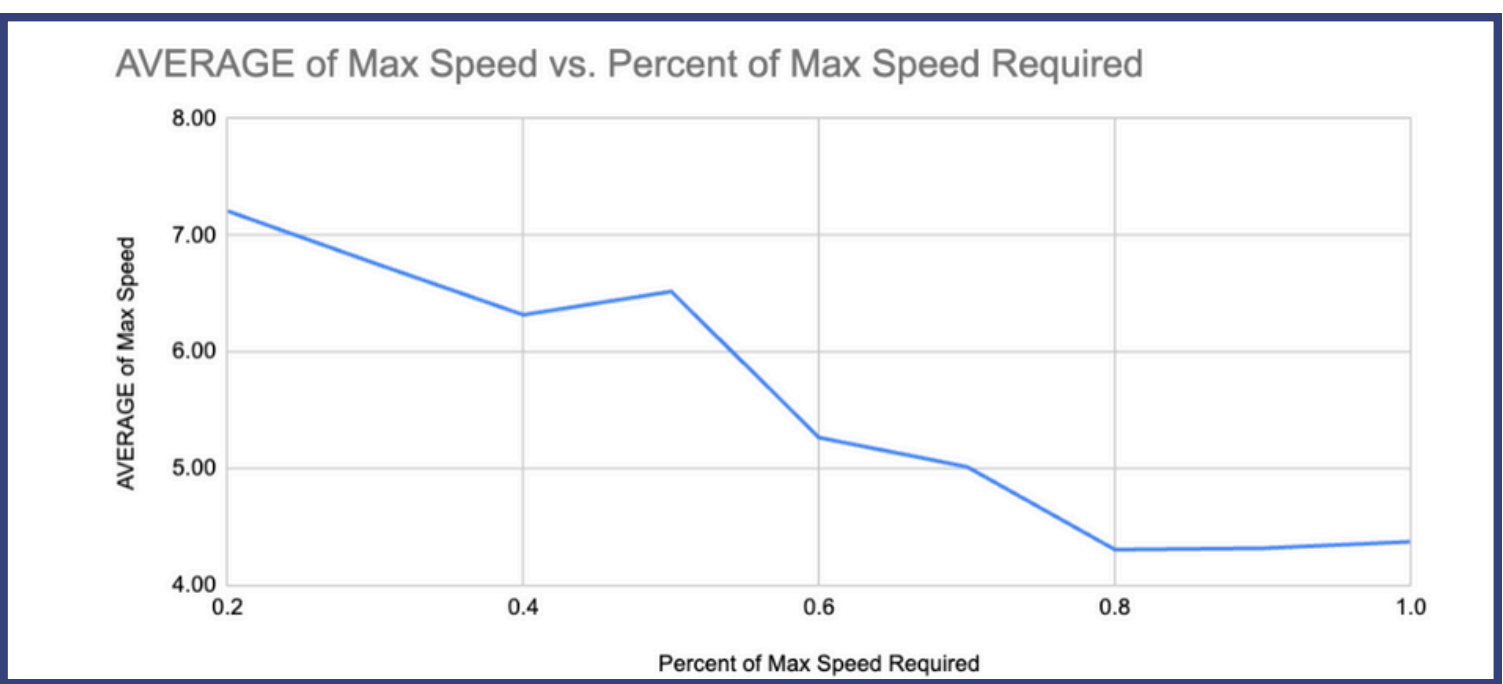
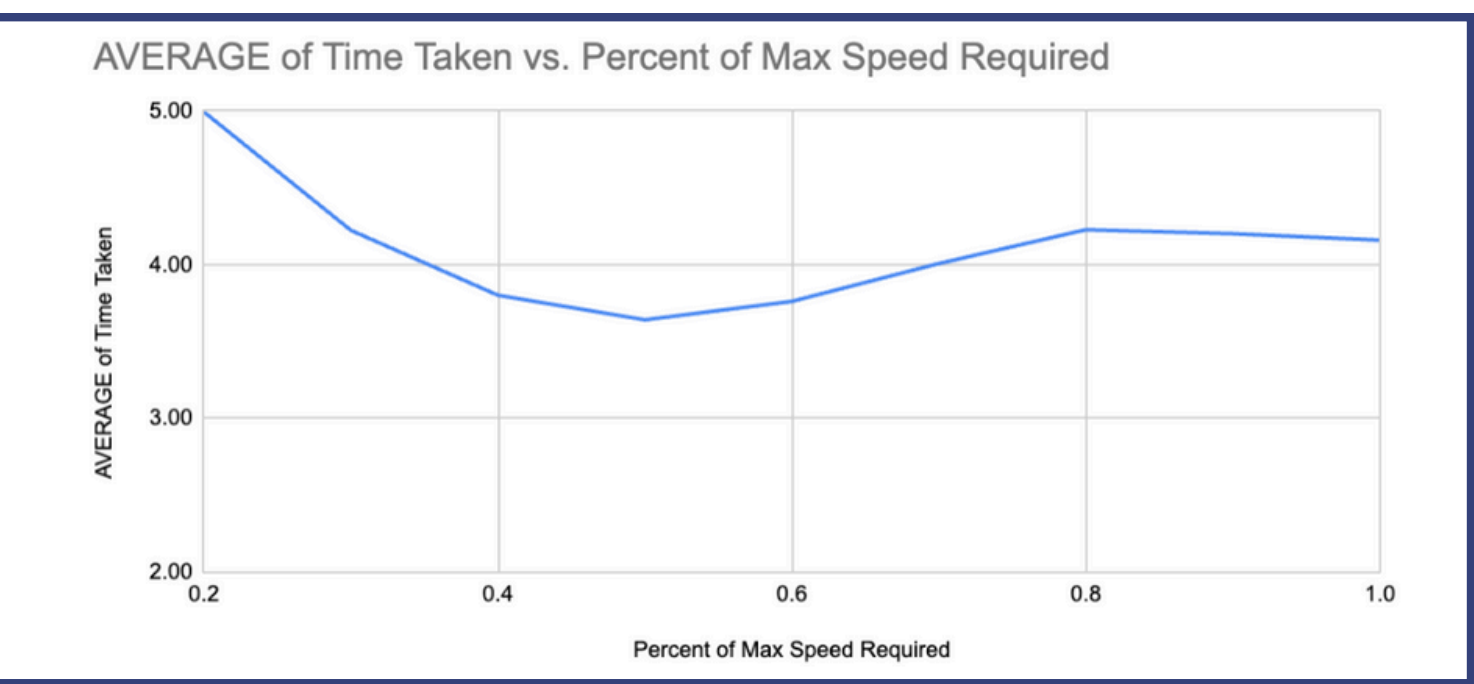


Figure 6-8: 12 ft. Motion Data

18 Feet			
Max Speed %	Avg Time Taken	Avg Max Speed	
0	9.999.00	5.05	
0.1	9.999.00	5.65	
0.2	4.99	7.21	
0.3	4.22	6.76	
0.4	3.80	6.32	
0.5	3.94	6.52	
0.6	3.76	5.27	
0.7	4.01	5.02	
0.8	4.23	4.31	
0.9	4.20	4.32	
1	4.16	4.374	

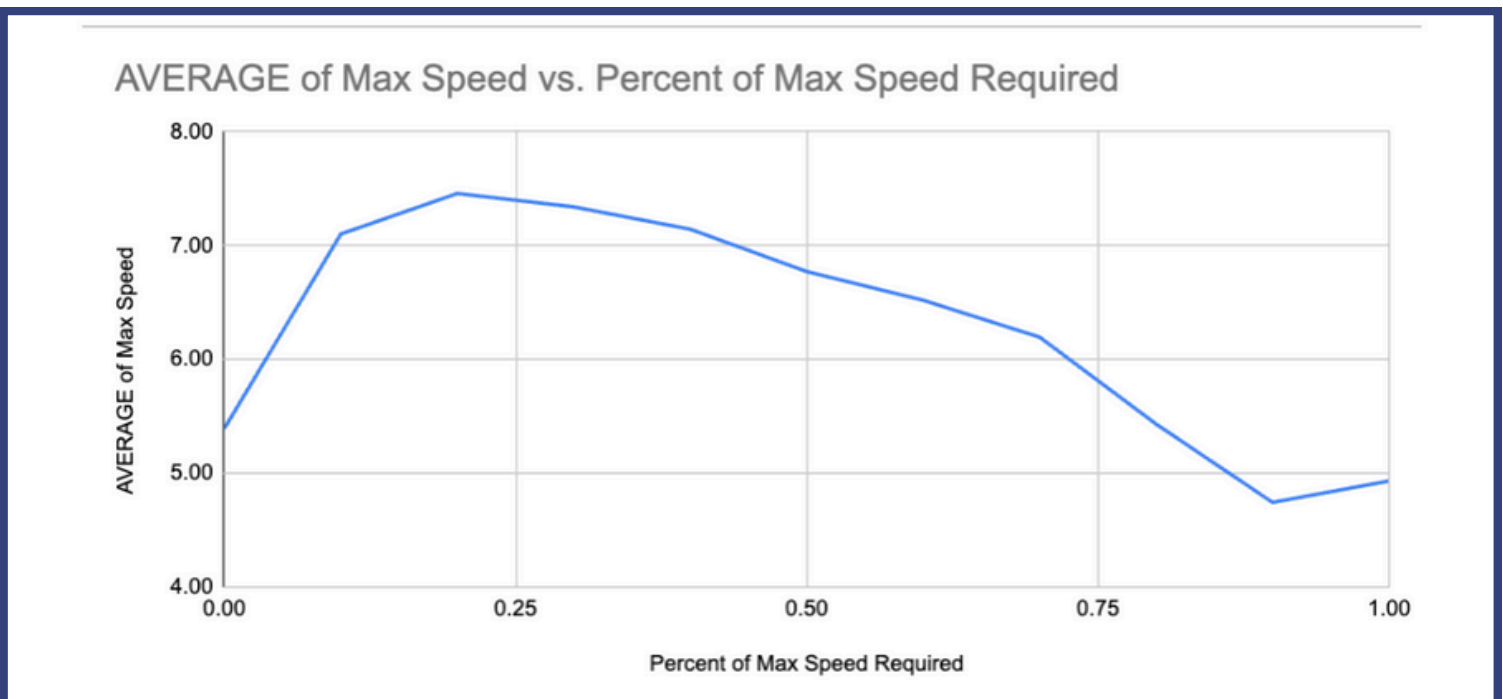
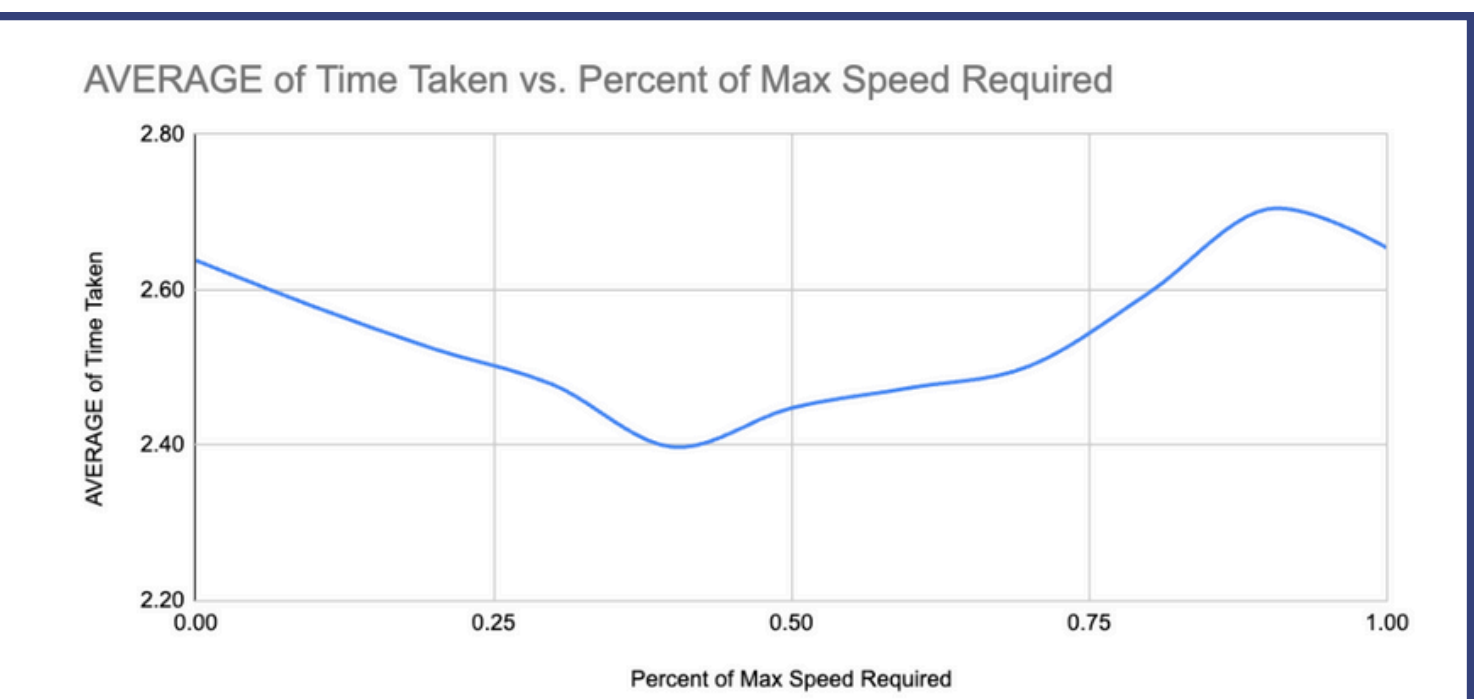


Figure 9-11: 18 ft. Motion Data

ANALYSIS

The existence of the CVT effect is validated by the velocity data. As the wheels angled inward, the maximum velocity of the robot increased substantially from **4.99 ft/s** (with no CVT) to **7.90 ft/s** (with CVT), representing a **63.07% increase in top speed**.

The time trials reveal a consistent pattern across both the 12-foot and 18-foot distances. As the target percentage of maximum speed for gear ratio change decreased from 100% to approximately 60%, travel times consistently improved. For 12 feet, time dropped from 2.65s to 2.40s (**9.4% improvement**), while for 18 feet, it fell from 4.16s to 3.64s (**12.5% improvement**). The data shows a clear "sweet spot" for implementing the gear ratio change. Between 40% and 60% of maximum speed, travel times remained relatively stable and optimal. However, when the gear ratio change was implemented below 40% or above 60% of maximum speed, performance deteriorated. Notably, when the maximum speed percentage was set below 20% for the 18-foot trials, the robot couldn't complete the course due to insufficient acceleration values.

Acceleration profiles reveal two distinct phases of motion. In the initial phase, rapid acceleration brings the robot close to its theoretical maximum speed, with acceleration values starting high and quickly diminishing to near zero. The second phase shows significantly lower acceleration values, resulting in gradually increasing speeds. This pattern indicates that **deploying the CVT effect is most effective after the robot has reached a stable speed**, allowing for smooth transition to higher maximum velocities, which backs the 40-60% value earlier found.

CONCLUSION

The research question addressed in this project was: How does the continuously variable transmission (CVT) effect in an omni-based swerve drive impact travel time, and what is the optimal speed percentage for dynamic gear ratio adjustments to maximize efficiency? The hypothesis proposed that reducing the gear ratio at 50% of top speed would minimize travel time.

The data collected **strongly supports this hypothesis**. The experimental results demonstrate first that the CVT effect, previously only theorized, can in fact be practically demonstrated. Second, it illustrates that implementing the gear ratio change when the robot reaches between 40% and 60% of its maximum speed produces the optimal travel times across both test distances. Not only that, but the results also uncover that the drivetrain exhibits two acceleration stages which must be smoothly transitioned between: an initial rapid increase of speed, followed by a slower, more gradual increase as the CVT is deployed.

Future work can be done to explore non-linear dynamic gear ratio changes which could optimize the motion even further. Additionally, experiments regarding multidirectional responses and on external factors should be conducted to truly determine an optimal CVT control.

These findings have significant implications for applications such as competitive robotics, autonomous vehicles, industrial automation, or transport systems where fast motion control is critical. This project overall demonstrates that an omni based swerve drive has potential for real world applications.

All images created by student researcher.